



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Mid Exam Summer 2025

CSE 2233/CSI 233: Theory of Computation/Theory of Computing

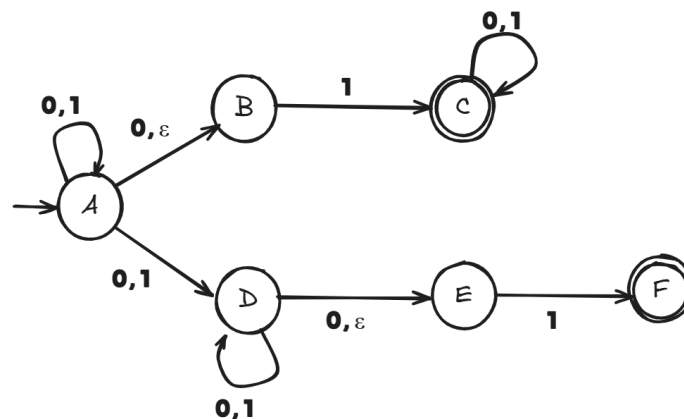
Total Marks: 30

Duration: 1 Hour 30 Minutes

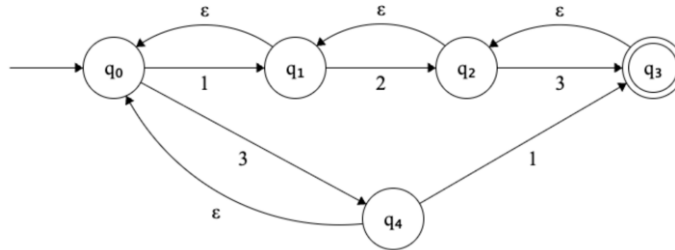
Answer all questions. Figures are in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. Design DFAs that accepts the following languages: 3x3
 - a) $L = \Sigma\Sigma \mid \Sigma = \{0,1,2\}$
 - b) $L = \{ w \mid w \text{ starts with an even number of } 0 \text{ and ends with } 112 \} \mid \Sigma = \{0,1,2\}$
 - c) $L = \{ w \mid w \text{ starts with } 011 \text{ and contains } 120 \text{ and ends with } 12 \} \mid \Sigma = \{0,1,2\}$
2. Design NFAs that accept the following languages: 3x3
 - a) $L = \{ w \mid w \text{ starts with } a \text{ and every } bb \text{ followed by at least one } c \} \mid \Sigma = \{a, b, c\}$
 - b) $L = \{ w \mid w \text{ starts and ends with different symbols when the length of the string is multiple of } 3 \} \mid \Sigma = \{a, b, c\}$
 - c) $L = \{ w \mid w \text{ starts with } ab, \text{ contains } abc \text{ or } aab \text{ and ends with } bb \text{ or } bc \} \mid \Sigma = \{a, b, c\}$
- 3 a) Consider the following ε -NFA, and show with the help of ε -NFA-tree whether the string “001” is accepted or not. 3



4. a) Convert the ε -NFA to equivalent DFA. Show both transition table and state diagram of the DFA. 6
Here, $\Sigma = \{1,2,3\}$:



- 5 Design Regular Expression for the following languages where $\Sigma = \{a,b\}$: 1x3
- $L = \{ w \mid w \text{ is of length at most } 3 \}$
 - $L = \{ w \mid w \text{ has **at least 4** characters and the 4th character is **b**.} \}$
 - $L = \{ w \mid w \text{ starts and ends with the same letter} \}$